

$$\lambda = 6.0 \text{ m}$$

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Ans: C

$$d \sin \theta = n \lambda$$

Violet light has a shorter wavelength than red light, thus it will be diffracted less from the normal than red light

$$\text{Second order of violet light: } d \sin \theta_v = 2 \times 350 \times 10^{-9} = 700 \times 10^{-9}$$

$$\text{First order of red light: } d \sin \theta_r = 1 \times 700 \times 10^{-9} = 700 \times 10^{-9}$$

For the same d, second order of violet light has the same angle of diffraction as that of red light of $700 \times 10^{-9} \text{ m}$

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Ans: B